

Experiment No. 14

Implementing Logic Gates Using Relays

Objective:

To understand and Implement Logic Gates using a set of Relays

Apparatus:

- Digital Multi-Meter (DMM)
- Connecting Wires
- Relays Trainer

Theory:

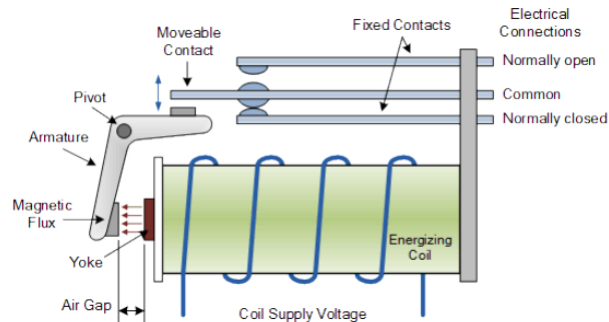
Relays

Relays are electro magnetically operated switches. An actuating current on a coil operates one or more galvanically separated contacts or load circuits. The electro mechanical relay is a remote-controlled switch capable of switching multiple circuits, either individually, simultaneously, or in sequence.

The primary functions of a relay are:

- The galvanic separation of the primary or actuating circuit and the load circuits
- Single input/multiple output capability
- Separation of different load circuits for multi-pole relays
- Separation of AC and DC circuits
- Interface between electronic and power circuits
- Multiple switching functions, e.g. delay, signal conditioning
- Amplifier

Typical applications for relays include laboratory instruments, telecommunication systems, computer interfaces, domestic appliances, air conditioning and heating, automotive electrics, traffic control, lighting control, building control, electric power control, business machines,



control of motors and solenoids, tooling machines, production and test equipment.

Electromechanical Relays:

Electromechanical relays are those where the switching element is a mechanical contact, actuated by an electromagnet. This is the most widely used type of relay design. The principal internal functions of an electromechanical relay are:

1. Conversion of electrical current (input, coil current) to a magnetic field
2. Conversion of the magnetic field into a mechanical force
3. This force operates the contacts (secondary side)
4. Contacts switch and conduct electrical current (output, load current)

The most important components of a Relay are:

Magnetic System (Primary Section)

- Coil (to generate the necessary magnetic field to actuate the armature and the contacts)
- Core (highly magnetic permeable - concentrates the magnetic field)
- Yoke (to establish the magnetic circuit)
- Armature (the moving part of the magnetic system which closes and opens the magnetic circuit and acts via an actuator or the moving relay contacts)

- Return spring (For quick return of the moving contact to normal condition on removal of the coil power)

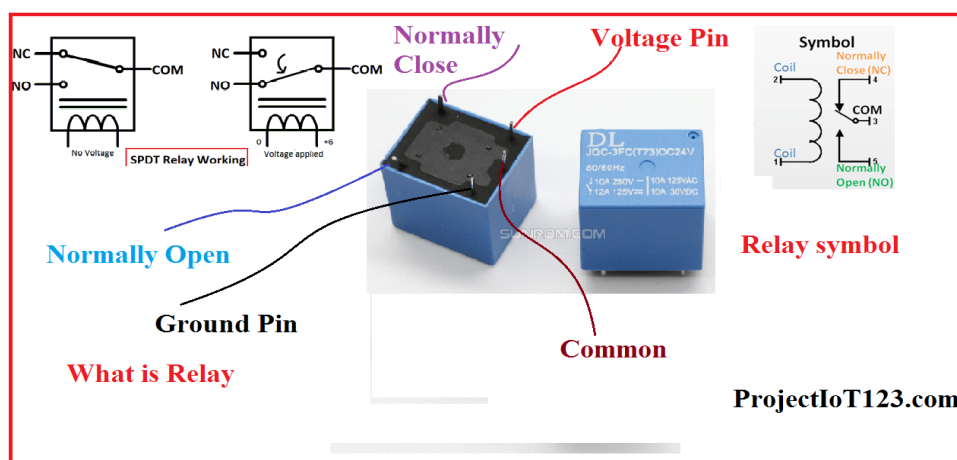
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Contact system (Secondary Section)

- Fixed contacts
- Moving contacts (contacts being moved by the magnetic system to switch the load circuit)
- Contact springs (holding the contacts but sufficiently flexible to allow the contacts to move)

Mechanical components

- Case & Base (to protect the relay against external influences and for protection against electric shock)
- Insulation within the relay to separate the primary circuit from the secondary side and to provide the required Actuator (used in some relay designs to translate the motion of the magnetic system to the contact system)
- Moving contacts must have insulation properties to isolate the primary circuit (coil, magnetic circuit) from the secondary side (contact system)
- Pins or terminals to connect between the contact system and the load



- Mounting devices such as sockets, solder hooks, PCB pins, brackets, captive screws

Different Types of Relays

Different types of relays are shown in the figure below:



Logic Gates

A logic gate is a device that acts as a building block for digital circuits. They perform basic logical functions that are fundamental to digital circuits. Most electronic devices we use today will have some form of logic gates in them. For example, logic gates can be used in technologies such as smartphones, tablets or within memory devices.

In a circuit, logic gates will make decisions based on a combination of digital signals coming from its inputs. Most logic gates have two inputs and one output. Logic gates are based on Boolean algebra. At any given moment, every terminal is in one of the two binary conditions, false or true. False represents 0, and true represents 1. Depending on the type of logic gate being used and the combination of inputs, the binary output will differ. A logic gate can be thought of like a light switch, wherein one position the output is off -- 0, and in another, it is on -- 1.

Basic Logic Gates

- **The AND gate** is so named because, if 0 is called "false" and 1 is called "true," the gate acts in the same way as the logical "and" operator. The following illustration and table show the circuit symbol and logic combinations for an AND gate. (In the symbol, the input terminals are at left and the output terminal is at right.) The output is "true" when both inputs are "true." Otherwise, the output is "false." In other words, the output is 1 only when both inputs one AND two are 1.

2 - input AND gate



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

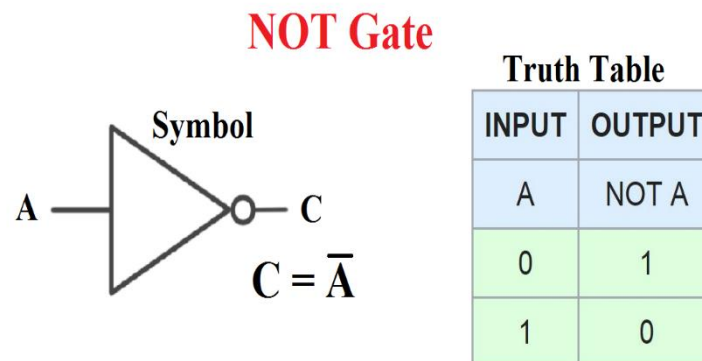
- **The OR gate** gets its name from the fact that it behaves after the fashion of the logical inclusive "or." The output is "true" if either or both of the inputs are "true." If both inputs are "false," then the output is "false." In other words, for the output to be 1, at least input one OR two must be 1.

2 - input OR gate



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

- A logical inverter, sometimes called a **NOT gate** to differentiate it from other types of electronic inverter devices, has only one input. It reverses the logic state. If the input is 1, then the output is 0. If the input is 0, then the output is 1.

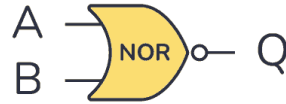


- **The NAND gate** operates as an AND gate followed by a NOT gate. It acts in the manner of the logical operation "and" followed by negation. The output is "false" if both inputs are "true." Otherwise, the output is "true."



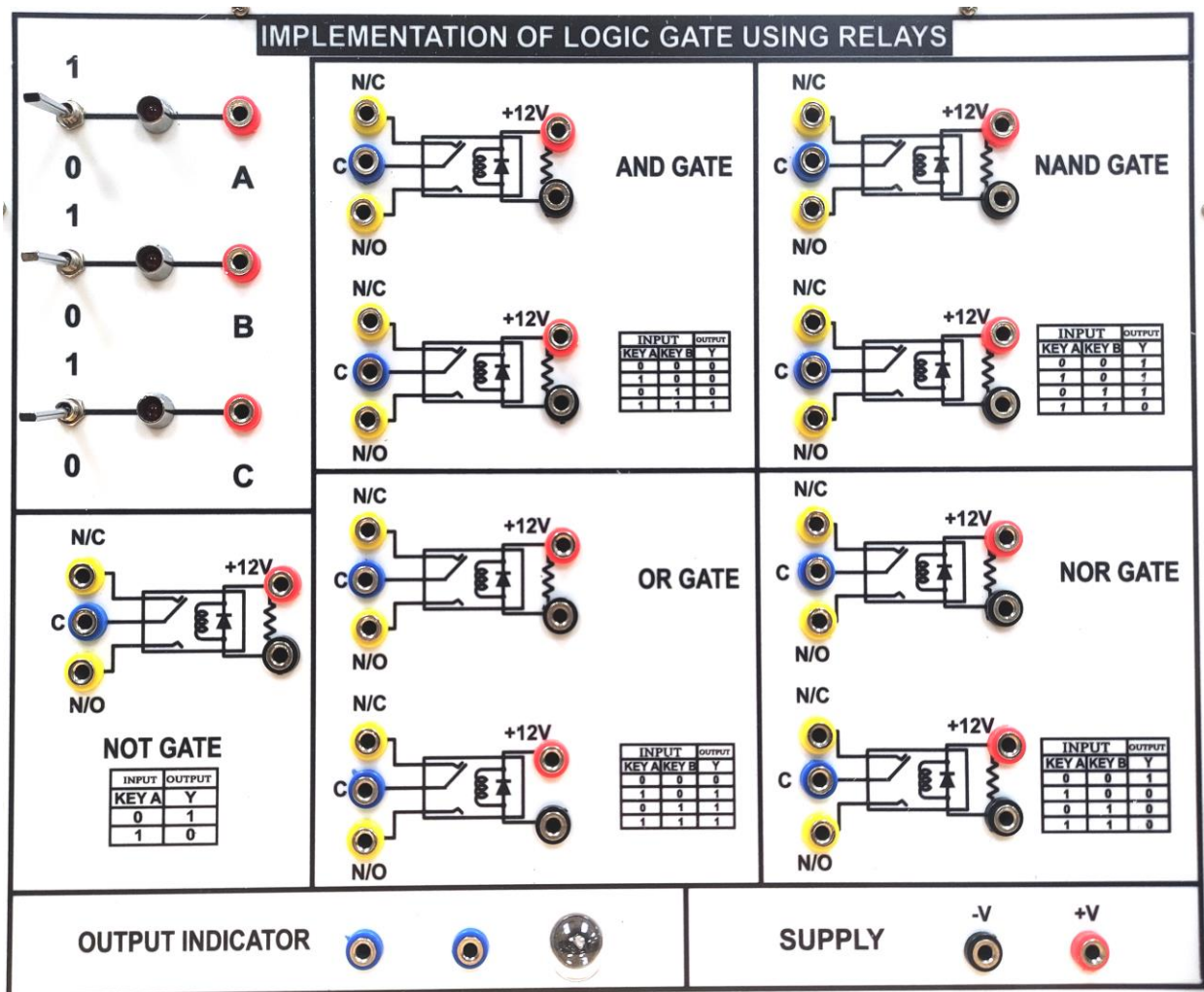
A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

- The **NOR gate** is a combination OR gate followed by an inverter. Its output is "true" if both inputs are "false." Otherwise, the output is "false."

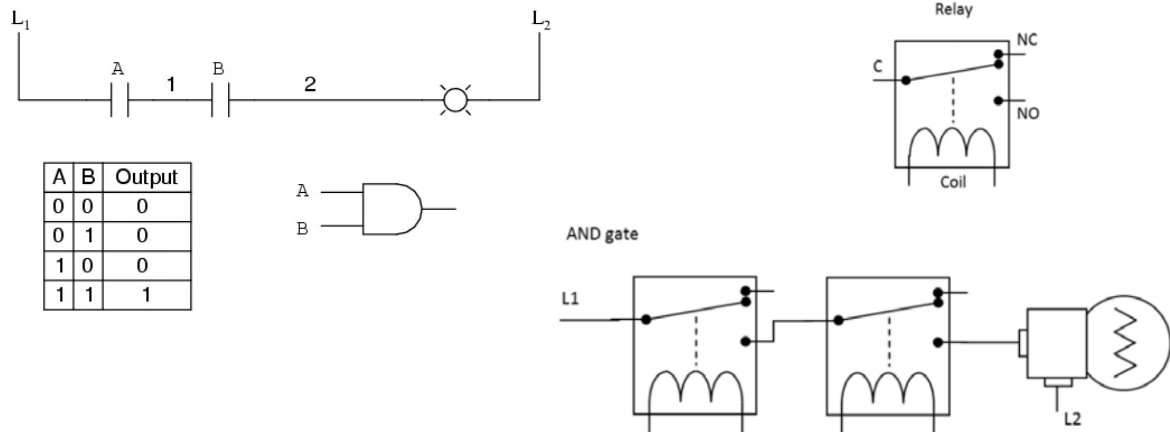


A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

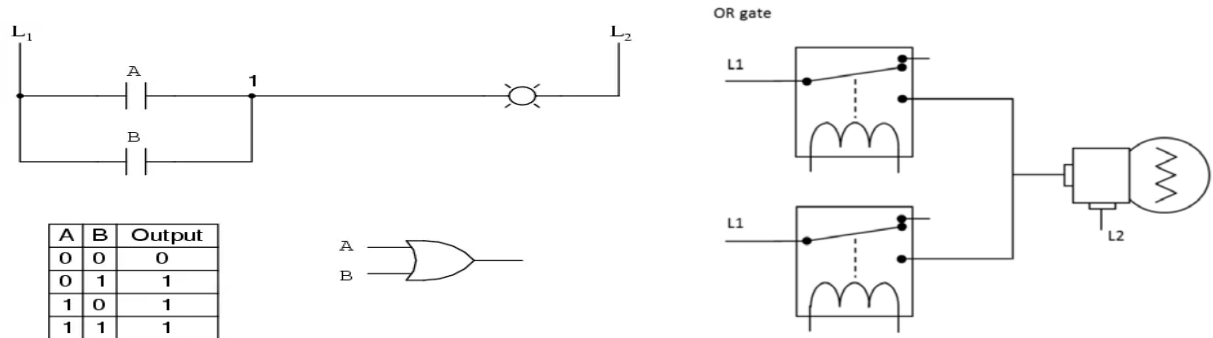
Circuit Diagram:



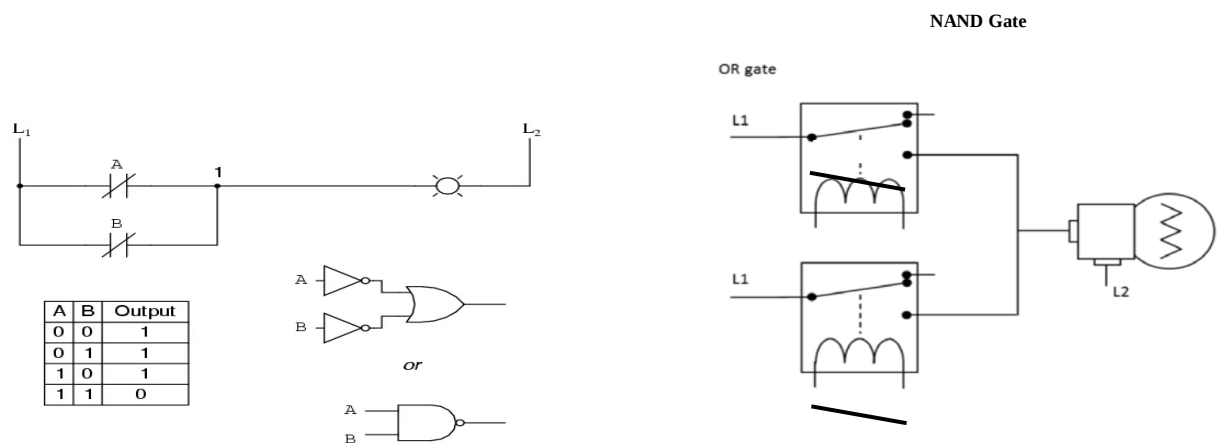
Implementing AND Gate Using Relay Logic:



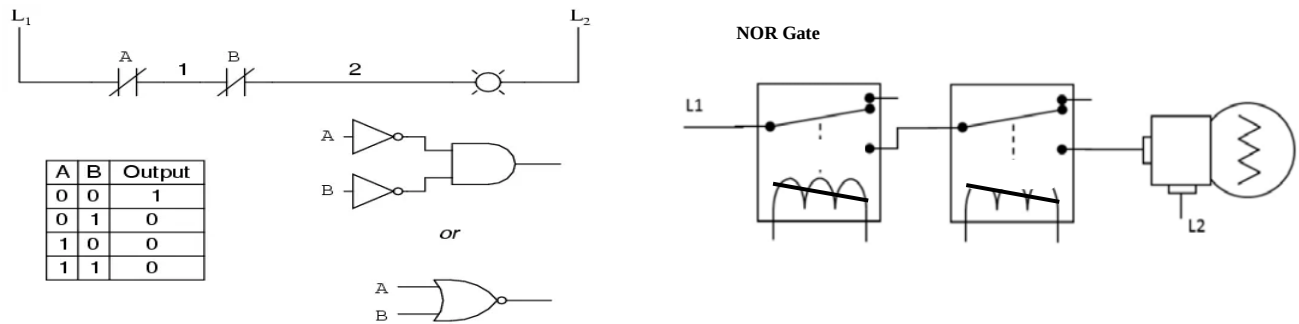
Implementing OR Gate using Relay Logic



Implementing NAND Gate using Relay Logic:



Implementing NOR Gate using Relay Logic



Activity No. 01

Implement AND Logic and NOT Logic Gate using Two Relays and record the Output, Also construct the Connection Diagram of the Circuit

A	B	OUT (AND Gate)	OUT (NOT Gate)
0	0		
0	1		
1	0		
1	1		

Connection Diagram:

Activity No. 02

Implement OR Logic Gate using Two Relays and record the Output, Also construct the Connection Diagram of the Circuit

A	B	OUT (OR Gate)
0	0	
0	1	
1	0	
1	1	

Connection Diagram

Activity No. 03

Implement NAND Logic Gate using Two Relays and record the Output, Also construct the Connection Diagram of the Circuit

A	B	OUT (NAND Gate)
0	0	
0	1	
1	0	
1	1	

Connection Diagram

Activity No. 04

Implement NOR Logic Gate using Two Relays and record the Output, Also construct the Connection Diagram of the Circuit

A	B	OUT (NOR Gate)
0	0	
0	1	
1	0	
1	1	

Connection Diagram

Lab Exercise and Summary

LABORATORY SKILLS ASSESMENT (Psychomotor)

Total Marks: 100

Criteria (Max Marks)	Level 1 $0\% \leq S < 50\%$	Level 2 $50\% \leq S < 70\%$	Level 3 $70\% \leq S < 90\%$	Level 4 $90\% \leq S \leq 100\%$	Score (S)
Procedural Awareness (20)	Selects inappropriate skills and/or strategies required by the task	Selects and applies appropriate skills and/or strategies required by the task with some errors	Selects and applies the appropriate strategies and/or skills specific to the task without significant errors	Selects and applies appropriate strategies and/or skills specific to the task without any error	
Practical Implementation (30)	Makes several critical errors in applying procedural knowledge of Implementing Logic Gates Using Relays	Makes few critical errors in applying procedural knowledge of Implementing Logic Gates Using Relays	Makes some non-critical errors in applying procedural knowledge of Implementing Logic Gates Using Relays	Applies the procedural knowledge of Implementing Logic Gates Using Relays in perfect ways	
Safety (10)	Requires constant reminders to follow safety procedures	Requires some reminders to follow safety procedures	Follows safety procedures with only minimal reminders	Routinely follows safety procedures	
Use of Tool/Equipment (20)	Uses tools, equipment and materials with limited competence	Uses tools, equipment and materials with some competence	Uses tools, equipment and materials with considerable competence	Uses tools, equipment and materials with a high degree of competence	
Participation to Achieve Group Goals (10)	Shows little commitment to group goals and fails to perform assigned roles	Demonstrates commitment to group goals, but has difficulty performing assigned roles	Demonstrates commitment to group goals and carries out assigned roles effectively	Actively helps to identify group goals and works effectively to meet them in all roles assumed	
Interpersonal Skills in Group Work (10)	Rarely interacts positively within a group, even with prompting	Interacts with other group members if prompted	Interacts with all group members spontaneously	Interacts positively with all group members and encourages such interaction in others	
Marks Obtained					

Instructor's Signature: _____

Date: _____

LABORATORY SKILLS ASSESMENT (Affective)

Total Marks: 40

Criteria (Max. Marks)	Level 1 $0\% \leq S < 50\%$	Level 2 $50\% \leq S < 70\%$	Level 3 $70\% \leq S < 90\%$	Level 4 $90\% \leq S \leq 100\%$	Score (S)
Introduction (5)	Very little background information provided or information is incorrect	Introduction is brief with some minor mistakes	Introduction is nearly complete, missing some minor points	Introduction complete and well-written; provides all necessary background principles for the experiment	
Procedure (5)	Many stages of the procedure are not entered on the lab report.	Many stages of the procedure are entered on the lab report.	The procedure could be more efficiently designed but most stages of the procedure are entered on the lab report.	The procedure is well designed and all stages of the procedure are entered on the lab report.	
Data Record (10)	Data is brief and missing significant pieces of information.	Data provides some significant information and has few critical mistakes.	Data is almost complete but has some minor mistakes.	Data is complete and relevant. Tables with units are provided. Graphs are labeled. All questions are answered correctly.	
Data Analysis (10)	Data is presented in very unclear manner. Error analysis is not included.	Data is presented in ways (charts, tables, graphs) that are not clear enough. Error analysis is included.	Data is presented in ways (charts, tables, graphs) that can be understood and interpreted. Error analysis is included.	Data are presented in ways (charts, tables, graphs) that best facilitate understanding and interpretation. Error analysis is included.	
Report Quality (10)	Report contains many errors.	Report is somewhat organized with some spelling or grammatical errors.	Report is well organized and cohesive but contains some grammatical errors.	Report is well organized and cohesive and contains no grammatical errors. Presentation seems polished.	
Marks Obtained					

LABORATORY SKILLS ASSESMENT (Cognitive)

Total Marks: 10

(If any)	
Marks Obtained	

Instructor's Signature: _____

Date: _____